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MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW TRU-6/A, 0-5000 PPH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of 0-5,000 pph rate of flow transmitter, designated TRU-6/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-J-25656	Jet Fuel, Grade JP-6
MIL-I-26299	Indicator, Rate of Flow EFU-2/A, 0-5,000 PPH, Integrally Lighted
MIL-I-27182	Indicator, Rate of Flow EFU-12A/A, 0-5,000 PPH, White Integral Lighting
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-M-45202	Magnesium Alloys, Anodic Treatment of

Naval Air Systems Command

SD-24	General Specifications for Design and Construction of Aircraft Weapons Systems

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (for Systems and Equipments)
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracing officer.)

3. REQUIREMENTS

3.1 Qualification. The transmitter furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitter except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

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3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.5 Toxic and corrosive fumes. The transmitter shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.6 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Titanium. Where high temperature and other stringent environmental conditions are a factor in the design of the transmitter, the use of titanium will be permitted, with the approval of the procuring activity where lightweight and high-strength materials are required. Use of lightweight materials and weight-saving designs shall be of major consideration and their use shall be investigated and exploited to the greatest possible extent. For Navy use, titanium (magnesium) shall conform to SD-24.

3.4 Design and construction. The transmitter shall be calibrated for rate of flow in pounds per hour of aviation fuel conforming to MIL-T-5624. The specific gravity of the fuel shall be 0.785 ± 0.005 at $15.6^{\circ}C$. A commercial fuel of higher flash point may be used.

3.4.1 Compatibility and interchangeability. The TRU-6/A transmitters shall be interchangeable with previously manufactured TRU-6/A transmitters and shall be completely compatible with either the EFU-2/A or the EFU-12A/A indicator conforming to MIL-I-26299 and MIL-I-27182, respectively.

3.4.2 The transmitter shall be designed to operate within the tolerances specified herein with an EFU-2/A or EFU-12/A indicator conforming to MIL-I-26299 and MIL-I-27182, respectively. The transmitter shall provide a synchro output to operate Clifton Precision Products Company' s type CRC-8-A-9 synchro torque receiver, or equal. For an increasing flow rate, the direction of rotation of the transmitter synchro shaft shall be counterclockwise as viewed from the shaft end.

3.4.2.1 When two EFU-2/A or two EFU-12A/A indicators are operated from one transmitter, the resulting errors of the indicator synchros shall not exceed 1° of arc over the errors resulting when only one indicator is used.

3.4.3 Power. The transmitter shall be designed to operate within the tolerances specified herein from 115V ±10 percent, 400 Hz±25 percent, single-phase power conforming in other characteristics to MIL-STD-704, except that the transmitter

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motor which receives regulated power from the EFU-2/A or EFU-12/A indicator shall be designed to operate from 65 to 126 V 400 ± 1 Hz power. The transmitter synchro shall operate with voltage surges to 195 ± 5V as specified in 4.6.10.

3.4.4 Range. The transmitter shall be designed to measure flow rates from 0 to 5,000 pph.

3.4.5 Mechanical stop. A mechanical stop shall be provided within the transmitter above the 5,000-pph point but below the zero point.

3.4.6 Pressure drop. The transmitter shall be so designed that when inoperative, the pressure drop will not be appreciably more under any fuel flow conditions than when it is operating normally.

3.4.7 Reliability and longevity

3.4.7.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785 (see 6.2.2).

3.4.7.2 Mean-time-between-failures (MTBF). The transmitter shall have a minimum acceptable MTBF of 1,000 hours demonstrated to a 90 percent confidence level.

3.4.7.3 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2,500 hours before wearout failures occur or the transmitter consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

3.4.8 Maintainability. The contractor shall establish a maintainability program in accordance with MIL-STD-470 which shall include specific design objectives for mean corrective maintenance downtime (M_{ct}), and maximum corrective maintenance downtime ($M_{max_{ct}}$), confidence and percentile levels, and equipment repair time. The following maintainability requirements shall be considered in the design of the transmitter:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

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d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility of all components requiring maintenance inspection, removal, or replacement.

3.5 Performance. The transmitter shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperatures: Operation at temperatures ranging from -54° to +190°C and exposure to temperatures ranging from -62° to +190°C with fuel ambient temperatures through the transmitter ranging from -54° to +71°C

b. Vibration: Vibration from 0.036 double amplitude to ± 20g at frequencies from 5 to 2,000 Hz

c. Altitude: Altitudes ranging from sea level to 100,000 feet

d. Shock: Shock forces up to 20g for 11 milliseconds

e. Salt fog: Salt fog for a period of 50 hours

f. Humidity: Relative humidity up to 95 percent

g. Fungus: Fungus growth as encountered in tropical climates

h. Voltage and frequency variation: Voltage frequency variations ranging from 104V 300 Hz to 126V 500 Hz

i. Dielectric strength: Application of 500V between any pin on the electrical receptacle and the transmitter case for 5 seconds

j. Voltage surge: Voltage surge up to 195V

k. Frequency cycling: Frequency cycling from 320 to 480 Hz

l. Response: Response in less than 4 seconds from zero to full scale.

3.5.1 Operating pressure. The transmitter shall be capable of operating in fuel systems with normal fuel pressure up to 1,000 psi, shall withstand, without damage, a static fuel pressure of 1,600 psi at both room temperature and -54°C, and shall withstand a static fuel pressure of 1,000 psi at a fuel temperature of 71°C when in an ambient temperature of 140°C.

3.5.2 Electromagnetic interference. The transmitter shall meet the radiated, conducted, and susceptibility electromagnetic interference requirements of MIL-STD-461 for class ID equipment except the frequency range shall be from 0.15 to 150 mc.

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3.5.3 Tolerances. The transmitter shall operate within the tolerances specified in tables I, II, and III.

TABLE I. Fuel Flow Transmitter Scale Error Tolerances

Fuel Flow PPH	Scale Error Tolerance PPH ±	
	Room Temperature and Substitute Fuel	High and Low Temperatures
200	20	30
500	20	30
1,000	15	25
1,500	15	35
2,000	20	40
2,500	25	40
3,000	30	40
3,500	35	45
4,000	40	50
4,500	45	55
5,000	50	60

TABLE II. Tests and Tolerances

Test		Tolerance PPH ±
Voltage and frequency variation system basis		1-1/2 percent of point above 500 pph
Position error		15
Friction		
Room temperature		10
High temperature		10
Low temperature		15
Vibration		
Pointer oscillation (double amplitude)	10g	20
Pointer variation	10g	20
Pointer oscillation (double amplitude)	20g	80
Pointer variation	20g	80

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TABLE III. Pressure Drop

Flow PPH	Maximum Pressure Drop PSI
2,000	0.6
4,000	2.2
Test to be conducted with end housing adapters supplied with basic transmitter	

3.5.4 Power consumption. Power consumption of the transmitter shall not exceed 20 va with 115 ± 1V 400 ± 2 Hz excitation when the transmitter is operating and providing an indication of 2,000 pph at room temperature conditions.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer' s part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer' s part numbers and changes thereto.

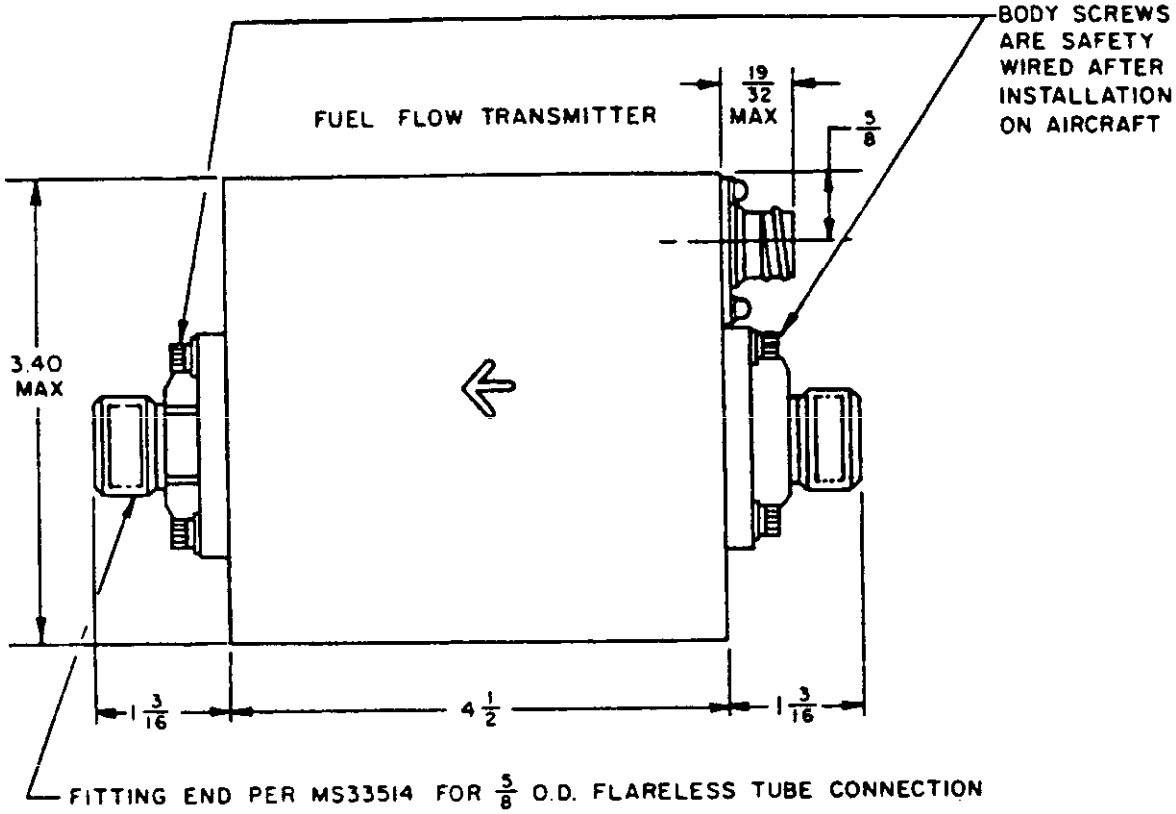
3.7 Mounting. The basic transmitter mounting bosses and spacing thereof shall be as shown on figures 1 and 2. The bosses for both ends of the transmitter shall be considered as part of the basic transmitter and shall be shipped with, but not assembled to, each transmitter. Protective leakproof fluid plates or caps shall be attached to the basic transmitter for shipment purposes.

3.8 Flow direction. The direction of fuel flow through the transmitter shall be permanently and legibly marked by an arrow, as shown on figure 1, on both the front and the back of the item.

3.9 Electrical receptacle. The transmitter shall be provided with a hermetically sealed electrical receptacle, Cannon part No. MC14H-10-6P, or equivalent. System connections shall be as shown on figure 3. Where practicable, the internal wiring shall be color coded and the pins shall be wired for the specified connections as follows:

Pin	Color	
A	Black/White	$(R_2)C_2$ lead of transmitter synchro rotor, low potential input of 115V 400-Hz regulated power to motor and external ground
B	Red/White	$(R_1)H_2$ lead of transmitter synchro rotor (115V 400-Hz input)
C	Blue	$(S_1)Y_2$ of transmitter synchro stator
D	Yellow	$(S_3)X_2$ of transmitter synchro stator
E	Black	$(S_2)Z_2$ of transmitter synchro stator (not connected to C_2)
F	Black	115V 400-Hz input from regulated power supply to motor lead.

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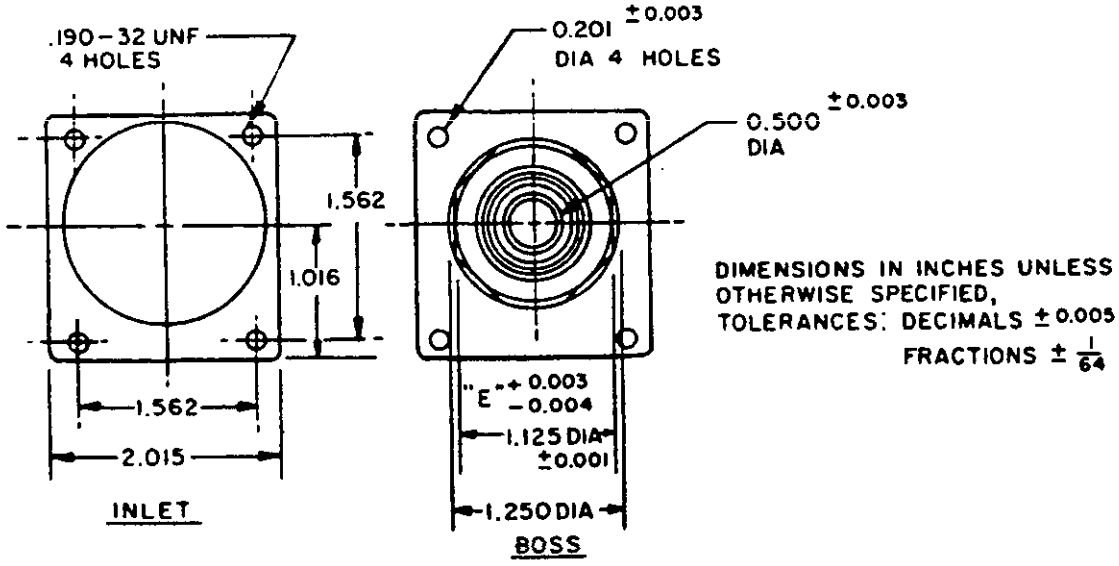
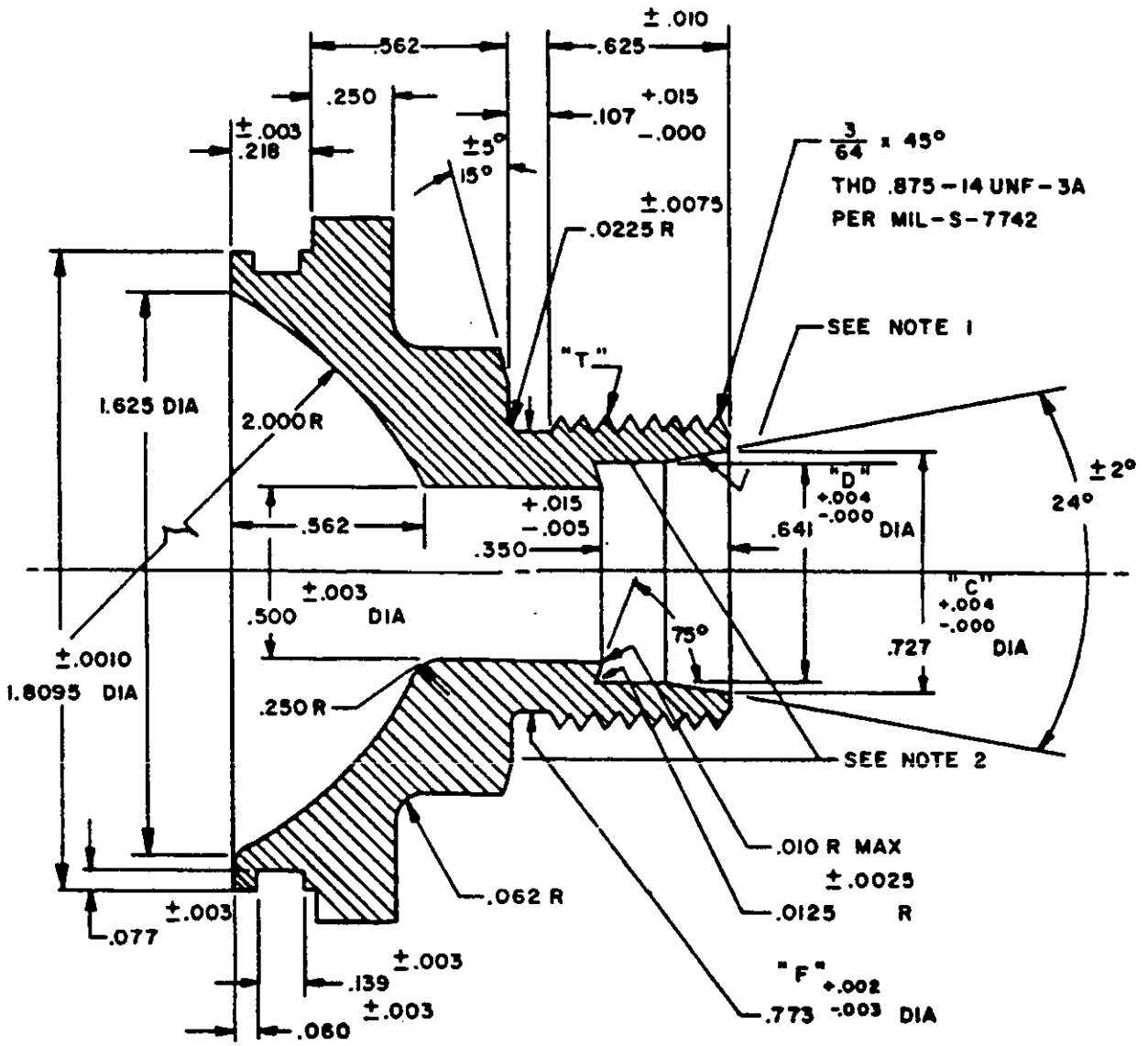


FIGURE 1. Basic Transmitter and Mounting

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NOTES:

1. This surface shall be free from burrs, longitudinal and spiral tool marks, and shall be smooth, except annular tool marks will be allowed to 63 microinches rms maximum. Styles G and E.

2. This surface shall be free from burrs, longitudinal and spiral tool marks, and shall be smooth, except that annular tool marks will be allowed to 100 microinches rms maximum.

3. Concentricity: D, C, and T (pitch diameter) shall be concentric to each other within 0.008 full indicator reading. F and T (pitch diameter) shall be concentric to each other within 0.005 full indicator reading.

Dimensions in inches. Unless otherwise specified, tolerance: Decimals ±0.016

FIGURE 2. Boss Detail

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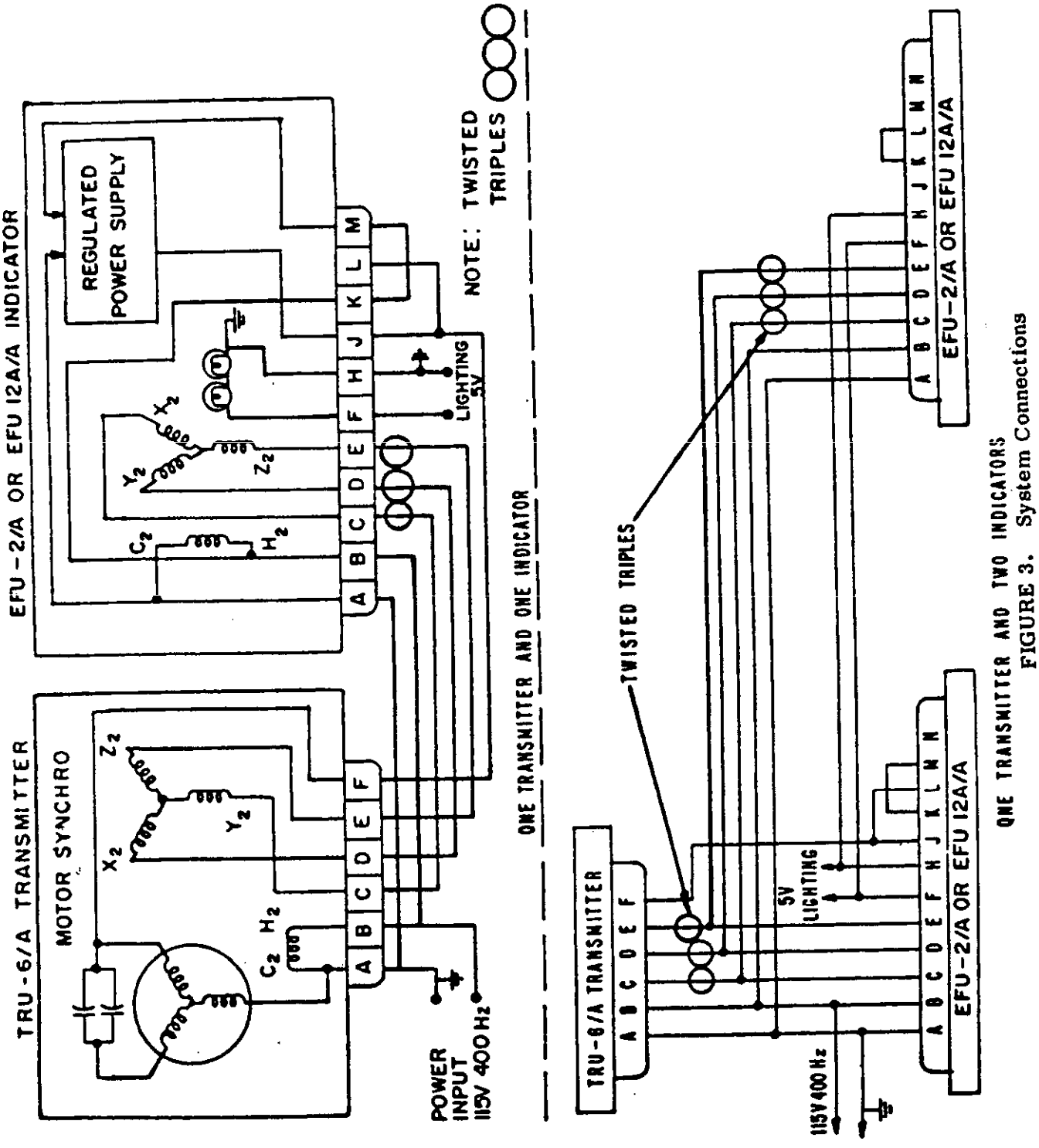


FIGURE 3. System Connections

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3.10 Electrical zero. For the purpose of this specification, the transmitter synchro rotor leads shall be designated C_2 and H_2. The stator leads shall be labeled X_2, Y_2, and Z_2. The subscripts indicate the requirement for a low null at electrical zero. Identification of the leads and the determination of electrical zero shall be as follows: Arbitrarily label one rotor lead as C_2, the low potential or ground lead, and label the other rotor lead H_2, the high potential lead. Arbitrarily label one stator lead Z_2 and connect Z_2 to C_2. Apply the rated voltage of the rotor between C_2 and H_2. Rotate the rotor to obtain the following conditions which indicate electrical zero, and label the other stator leads accordingly:

a. The voltage between H_2 and one of the unlabeled stator leads will be a maximum and the voltage between the unlabeled stator leads will be a minimum, or at null

b. The voltage between one of the unlabeled stator leads and the Z_2 lead shall increase in phase with the excitation voltage C_2 to H_2 before decreasing when the rotor is rotated in a clockwise direction from electrical zero as viewed from the shaft end. Label this lead X_2. Label the remaining stator lead Y_2. At electrical zero, the voltages X_2 to Z_2 and Y_2 to Z_2 shall be approximately in phase with excitation voltage C_2 to H_2. Schematically, the stator leads will be designated X_2, Y_2, Z_2 in a counterclockwise direction.

3.11 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall conform to MIL-B-27497.

3.12 Dimensions. The outline dimensions shall conform to figure 1.

3.13 Weight. The weight of the complete transmitter shall not exceed 5 pounds.

3.14 Contaminated fuel. The transmitter shall operate as specified herein with contaminated fuel containing foreign particles of the sizes shown in table IV.

TABLE IV. Contaminated Fuel Particle Size

Foreign Particle Size (Microns)	Percent of Total Weight
0-5	39 ±2
5-10	18 ±3
10-20	16 ±3
20-40	18 ±3
over 40	9 ±3
Through a 200 mesh screen	100 percent

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3.15 Caps. The fuel-line connections shall be provided with protective shipping caps or plugs to prevent foreign material from entering the transmitter.

3.16 Finishes and protective coatings

3.16.1 Alloy parts. Where practicable, aluminum or titanium alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts, and small areas that must be machined after finishing need not be covered with a protective film.

3.16.2 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable paint base, or for castings containing nonaluminum or titanium alloy inserts, a chemical treatment in accordance with MIL-C-5541 may be used in lieu of an anodic film for alloys that do not anodize satisfactorily.

3.16.3 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

3.16.4 Magnesium parts. Magnesium alloy parts shall be covered with an anodic film in accordance with MIL-M-45202, HAE process.

3.17 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17.1 Nameplate. A nameplate shall be provided in accordance with MIL-STD-130. Metal decalcomanias shall not be used.

3.17.2 Schematic. An electric schematic showing the function of each connector pin shall be permanently attached to the body of the transmitter.

3.18 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Test power. The supply power shall be single-phase ac, 115V ±2 percent, and 400 Hz controlled to less than ±1 Hz and of known value.

4.3.2 Test fuel. Unless otherwise specified, the fuel used for calibration in testing shall be JP-4 conforming to MIL-T-5624. The specific gravity of the fuel shall be 0.785 ±0.005 at 15.6°C. (Note: An approved commercial equivalent with higher flash point may be used.)

4.3.3 Test position. Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o' clock position when viewed from the inlet end.

4.3.4 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.5 Vibration. Unless otherwise specified, the transmitter shall be vibrated before a test reading is taken. An electric vibrator, set at 3,600 Hz with a maximum amplitude of 0.002 inch and attached to the flow pipe may be used.

4.3.6 Test equipment. Unless otherwise specified, the transmitter shall be tested while connected to a master indicator and master power supply. The master power supply shall be such that it will have no frequency error, or if a frequency error should exist (not exceeding 1 Hz at 400 Hz), it shall be subtracted from the test data. All flow tests shall be performed on a primary flow stand which measures liquid mass flow rate directly in terms of weight and time.

4.3.7 Tolerances. Test tolerances shall be as specified in tables I, II, and III.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of at least fourteen transmitters representative of the production equipment. Eleven of the samples shall be subjected to the individual, relaibility (see 4.6.29.1) and longevity tests only and

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the other three samples shall be subjected to all the tests except reliability and longevity. The samples shall be identified with the manufacturer' s part number and such other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.1).

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6. Unless otherwise specified, tests shall be conducted in the order determined by the qualifying activity.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Scale error

d. Friction error

e. Leakage

f. Response time

g. Burn-in.

4.5.2 Sampling tests. A minimum of 3 transmitters shall be selected at random from each 100 or fraction thereof produced on the contract and subjected to the following tests as described under 4.6:

a. Individual tests

b. Position error

c. Voltage and frequency variation

d. Voltage surge

e. Power consumption

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f. Low temperature

g. High temperature

h. Vibration

i. Pressure drop.

4.5.2.1 Reliability sampling. For reliability sampling, a lot shall consist of all transmitters produced on a specific contract. A minimum of 5 transmitters shall be selected at random from each 100 or fraction thereof produced on the contract and subjected to the test specified in 4.6.29.2.

4.5.2.2 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, the tests specified by the procuring activity shall be repeated.

4.5.2.2.1 Reconditioning of tested transmitters. Unless otherwise specified (see 6.2), transmitters that have been subjected to the sampling tests shall be reconditioned by the contractor by replacing all worn or damaged parts. After reworking, the contractor shall resubmit the transmitters for acceptance.

4.5.2.3 Individual tests may continue. Individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted (see 6.2). If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish, workmanship, and marking. Each unit shall be weighed.

4.6.2 Dielectric strength. A potential of 500V at a commercial frequency shall be applied between any pin and the transmitter case for a minimum of 5 seconds, There shall be no insulation breakdown as a result of this test.

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4.6.3 Scale error. The scale error of each transmitter shall be determined at the test points specified in table I at room temperature. The scale error shall be determined for both increasing and decreasing flow rates. Each test point in a set of readings shall be consistently approached from the same direction. A minimum of two sets of readings shall be taken at each test point. A set of readings shall consist of two flowstand weigh cycle times averaged to obtain average flow rates for two independent runs. The values for both runs shall be reported. If the individual readings for one of the runs exceeds or is within 5 percent of the specification limit, a second set of readings shall be taken. Scale error shall be determined while the transmitter is gently vibrated by an electromechanical vibrator mounted on the flow piping. Scale error tests conducted at room, low, and high temperatures, and substitute fuel operating conditions shall be in accordance with this procedure. Test tolerances shall be in accordance with table I.

4.6.4 Friction error. Friction error shall be determined at 25, 50, and 90 percent of full scale. The difference in readings taken before vibration and after vibration shall not exceed the values listed in table II. This test shall be combined with the scale error test.

4.6.5 Leakage. The transmitter shall be tested for leakage as follows: A nonshock hydraulic pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. A pressure of 1,600 psi shall then be rapidly applied five times at room temperature. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for a period of 5 minutes as checked on a pressure gage readable to 10 psi or by visual leakage of a colored test fluid. The transmitter temperature shall then be reduced to $-54 \pm 2^{\circ}C$ by submersion in a cold liquid bath, already at this temperature, for a period of not less than 15 minutes. While still at this temperature, the test pressure shall be rapidly applied five times. After the last application of pressure, there shall be no evidence of leakage for a period of 5 minutes as detected by the same means used for the room temperature portion of this test.

4.6.6 Response time. The time required for the transmitter output signal to come within 1 percent of the final steady state signal shall not exceed 4.0 seconds when a step change in flow rate is applied. The flow rate shall be changed from approximately 2 percent of full scale to a full scale indication by the activation of a quick opening valve. The timer will be started simultaneously with the activation of the valve.

4.6.7 Burn-in. The transmitter shall be placed in a chamber with the temperature maintained at 100°C for a minimum of 4 hours. The temperature shall then be lowered and maintained at 50°C for 20 hours with fuel flowing equivalent to a half-scale indication. At least the first 8 hours of this test shall be performed during normal working hours. Observations shall be made and recorded at least once every 2 hours throughout the 8-hour period. At no time shall the transmitter output change by more than one-half the tolerance specified for this test point in the applicable tolerance table. Pointer oscillation or erratic movement shall be cause for rejection. The transmitter shall be returned to room temperature for a minimum of 2 hours after which it

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shall meet the test specified in 4.6.3. If the burn-in test is conducted other than as specified herein, the test shall be subject to the approval of the procuring activity.

4.6.8 Position error. Position error shall be determined at 50 percent of full scale Readings shall be taken in the normal position and with the transmitter turned 90° in any direction. The reading taken in any position shall not differ from the reading taken in the normal position by more than the values specified in table II.

4.6.9 Voltage and frequency variation. At pins B and A, the applied voltage and frequency shall be set at the following points:

	Voltage	Frequency (Hz)
Normal condition	115	400
	104.0	300
	104.0	500
	126.0	300
	126.0	500

This test shall be conducted at 25, 50, and 90 percent of full scale flow rates. Readings shall be taken at the 115V, 400 Hz normal condition and at the conditions specified herein. The reading taken at each of the above conditions shall not differ from the reading taken at the normal condition in excess of the value shown in table II The transmitter shall be operated at least 1/2 hour at each of the specified conditions before any readings are taken. These tests shall be run with a master indicator and a master power supply (see 4.3.6).

4.6.9.1 The test specified in 4.6.9 shall be repeated with power applied to pins A and F and with the voltage and frequency set as follows:

	Voltage	Frequency (Hz)
Normal condition	115	400
	65	400
	126	400

4.6.10 Voltage surge. A master indicator need not be used during the surge application of this test. A synchro receiver, or simulated receiver, compatible with the transmitter will be adequate. A master power supply must be used. The test shall be made at approximately 50 percent of full scale. Power shall be applied at 115 ±1V, 400 ±2 Hz for 1 hour. At the end of this operating period, the power input shall be adjusted to 115 ±1V, 320 ±2Hz. A voltage surge shall then be applied to provide an rms peak voltage of 160 ±5V and return to 115 ±1V. The duration of the surge including rise and decay time shall be 0.09 ±.005 second. The power input shall then be adjusted to 115 ±1V, 480 ±2 Hz and a voltage surge applied to provide an rms peak voltage of 195 ±5V and return to 115 ±1V. The duration of this surge including rise

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and decay time shall be 0.07 ±.005 second. At the conclusion of this procedure, a scale error test at room temperature shall be conducted. The errors shall not exceed the room temperature tolerances of table I.

4.6.11 Power consumption. With 115V 400 Hz and power from the indicator to transmitter motor applied to the transmitter, power consumption of the transmitter shall not exceed 20 va at a flow rate of 2,000 pph.

4.6.12 Low temperature. The transmitter shall be placed in a test chamber and maintained at a temperature of -54° ±2°C for a period of 4 hours. At the end of this period and while still at the specified low temperature, the tramsmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables I and II, respectively.

4.6.13 High temperature. After the chamber and transmitter have been raised to a temperature of 121°±2°C, the transmitter shall be operated at that temperature for 6 hours. While at the specified high temperature, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables I and II, respectively.

4.6.14 Vibration. The vibration test shall be conducted in accordance with MIL-STD-810, method 514, equipment category (b), curve H, at room temperature with fuel flowing at approximately 50 percent of full scale. Pointer oscillation and variation shall not exceed the tolerance specified in table II for 10g. At the completion of this test, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. Scale and friction errors shall not exceed the tolerances specified in tables I and II, respectively.

4.6.14.1 Vibration endurance. The transmitter shall be subjected to an endurance vibration cycling test for 40 hours in each of the three mutually perpendicular axes. The vibration spectrum shall be in accordance with MIL-STD-810, method 514, equipment category (b), curve L, and the duration shall be in accordance with the figure titled "Logarmithmic cycling rate". During this test, the scale error test shall be conducted at 10, 20, 40, 60, and 100 percent of full scale. Scale error shall not exceed the tolerance specified in table II for vibration at 20g. After completion of this test, the scale error test shall again be conducted. Errors shall not exceed the tolerances specified in table I for high and low temperatures. The transmitter shall be completely disassembled and examined for damage or wear that might result in failure during the design life of the instrument.

4.6.15 Pressure drop. The pressure drop across the transmitter shall be determined at the flow rates specified in table III. The pressure drop shall be recorded with the transmitter both energized and unenergized. The pressure drop shall be obtained by sensing the live pressure at the inlet and outlet ends of the flowmeter and then obtaining the difference between the two readings. The pressure drop shall not exceed the values specified in table III.

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4.6.16 Low temperature exposure. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -62°C for a period of 72 hours at which time the equipment shall be inspected for deterioration. The temperature of the chamber shall then be raised to -54°C and maintained for an additional 24-hour period. At the end of the 24-hour period and while still at -54°C, the transmitter shall be subjected to the scale error and friction error tests. The transmitter shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the scale error and friction error tests at room temperature. The transmitter shall be examined and there shall be no deterioration of nor damage to the transmitter as a result of the extreme temperature exposure. The errors shall not exceed those specified in tables I and II.

4.6.17 Temperature-altitude. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -54° ±2°C and at a pressure equivalent to 100,000 feet for a period of 4 hours. The transmitter shall be operated with no fuel flowing. Immediately after the 4-hour period at the specified temperature and pressure, the transmitter shall be returned to normal room conditions and the scale error and friction error tests shall be conducted. The errors shall not exceed those specified in tables I and II for room-temperature conditions. This test may be combined with the test specified in 4.6.16. There shall be no damage to nor deterioration of the transmitter as a result of this test.

4.6.18 High temperature exposure. While inoperative, the transmitter shall be subjected to a temperature of 121°±2°C for a period of 4 hours. The temperature shall then be raised to 190°±2°C for a period of 1 hour. This cycle shall be repeated for a total of 48 hours at 121°±2°C and 12 hours at 190°±2°C during which fluid may be inside the transmitter but not flowing. The transmitter shall be energized and operated for the last two temperature cycles after which it shall be operated at 140° ±2°C for 4 hours at 1,000 pph with fuel flowing. The temperature of the fuel shall be 70°±2°C. The transmitter shall then be immediately tested for scale error and friction at 121°±2°C ambient air and the fuel at 50°±2°C. The tolerances shall not exceed those specified in tables I and II. The transmitter shall be examined and there shall be no evidence of failure.

4.6.19 Pressure leakage-high temperature. The transmitter shall be subjected to a pressure leakage at high temperature test as follows: A nonshock pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. The transmitter shall be heated in a 140°C chamber for 1/2 hour and a pressure of 1,000 psi shall then be applied rapidly five times. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for 5 minutes as checked by a pressure gage readable to 10 psi or by visual leakage of a colored test fluid.

4.6.20 Humidity. The transmitter shall be subjected to a humidity test in accordance with MIL-STD-810, method 507 after which it shall be subjected to the tests specified in 4.6.3 and 4.6.4.

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4.6.21 Fungus. The transmitter shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I. At the conclusion of this test, the transmitter shall be visually examined. There shall be no deterioration or corrosion of internal components as a result of this test.

4.6.22 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I, for a period of 50 hours. The scale error and friction error tests shall be conducted before and after this test. The errors shall not exceed the low and high temperature tolerances. The transmitter shall also pass the test specified in 4.6.2.

4.6.23 Electromagnetic interference. The transmitter shall be subjected to electromagnetic conducted, radiated, and susceptibility tests in accordance with MIL-STD-462 for class ID equipment except the frequency range shall be from 0.15 to 150 mc.

4.6.24 Shock. The shock test shall be conducted in accordance with MIL-STD-810, method 516, procedure I.

4.6.25 Contaminated fuel operation. Test fuel containing dirt to the extent of 800 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for 1 hour. Scale error and friction error tests shall then be conducted using the contaminated fuel. Operation shall be satisfactory and within the room temperature tolerances. The foreign matter shall consist of not less than 68 percent SiO_2 and shall have a particle-size analysis as shown in table IV.

4.6.26 Substitute fuel. This test shall be conducted using JP-6 fuel conforming to MIL-J-25656 and at room operating temperatures. The scale errors shall not exceed the tolerances specified in table I for high and low temperature.

4.6.27 Frequency cycling. The master indicator need not be used for the cycling portion of this test. A second harmonic, or simulated receiver, compatible with the transmitter will be adequate. A master power supply shall be used. The voltage shall be adjusted to 115 ±1V and the frequency cycled from 320 ±2 to 480 ±2 Hz in accordance with the following schedule with approximately half-scale flow. This constitutes one cycle for a total time of 100 seconds.

a. Start cycle at 320 Hz, increase frequency to 480 Hz in 38 seconds linearly with time

b. Return frequency to 320 Hz in 2 seconds or less

c. Increase frequency to 480 Hz in 58 seconds linearly with time

d. Return frequency to 320 Hz in 2 seconds or less.

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The cycle shall be repeated continuously for 10 hours at the rate of 36 cycles per hour. Power shall then be removed for a maximum of 2 hours and reapplied for 1 hour at 120 ±1V 320 ±2 Hz. Immediately following the 1 hour of operation at 120V 320 Hz, the transmitter shall be frequency cycled for 1 additional hour on the schedule indicated above, except that the frequency range shall be increased from 300 to 500 Hz. The transmitter shall then be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerances specified in table I for room temperature.

4.6.28 Compatibility. The transmitter shall be subjected to the tests specified in 4.6.3, 4.6.9, and any other tests necessary to determine compliance with 3.4.1, 3.4.2, 3.4.3, 3.5.4, and 3.8.

4.6.29 Reliability

4.6.29.1 Qualification. The 11 transmitters specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level J, which shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level.

4.6.29.1.1 Test equipment. The test equipment and fixtures shall include provisions for transmitter operation, temperature cycling, and vibration.

4.6.29.1.2 Operation. During this test, fuel at room temperature shall be flowing through the transmitter at a rate from 5 to 90 ±5 percent of full scale and return at a rate of 2 ±1 cpm.

4.6.29.1.3 Temperature cycling. Unless otherwise specified or amended by the pro curing activity due to the results of the thermal analysis, the temperature of the chamber in which the transmitters are tested shall be cycled as follows:

a. Starting at room temperature, decrease temperature to -54°C and stabilize for 1 hour with transmitters unenergized

b. Turn transmitters on, raise temperature to 125°C, and operate transmitters for 3 hours

c. Turn transmitters off and lower chamber temperature to -54°C for 1 hour

d. Repeat steps b and c until reliability test requirement has been met. This schedule provides six 4-hour test cycles for each 24-hour period. In each 24-hour period the operational test time is 18 hours.

4.6.29.1.4 Vibration cycling. Throughout this test, the transmitter shall be vibrated for a period of 10 minutes during each hour of On time at any nonresonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2g 10 percent. During this test, the transmitter motor shall undergo voltage cycling from 65V 400 Hz to 126V 400 Hz linearly in time at a rate of 1 cycle every 4 ±1 minutes during operational On time starting with 115V 400 Hz.

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4.6.29.1.5 Performance check. At each 200-hour interval and at the end of this test, the transmitters shall be subjected to the scale error, friction, and response time tests. During each scale error test, if any two test points exceed twice the allowable tolerances for high and low temperatures specified in tables I and II, one failure shall be charged against this test.

4.6.29.1.6 Failure criteria. For an accept decision, each transmitter shall be operated at least 600 hours. If a transmitter fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the transmitters still under test. The procuring activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, test plan I.

4.6.29.1.7 Reporting. Reporting shall be submitted in accordance with MIL-STD-781. The reliability test log and failure summary and analysis shall be included in the final test report (see 6.2.1).

4.6.29.2 Quality conformance. The sample transmitters specified in 4.5.2.1 shall be subjected to a reliability test in accordance with the "Reduced Testing" requirements of MIL-STD-781, test level J, and as specified under 4.6.29.1 which shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level. Accept-reject criteria shall be in accordance with test plan V except the discrimination ratio shall be 1.5. When "Tightened Testing" is required by the qualifying activity, acceptance shall be in accordance with test plan I or III as determined by that activity. Results of this test shall be included in the test report specified in 4.6.29.1.7.

4.6.30 Longevity. Testing on 2 of the 11 reliability test samples shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII, until 2,500 hours have accumulated on each transmitter.

4.6.31 Maintainability. The contractor shall submit his maintainability program to the procuring activity for review and approval to determine that the requirements specified in 3.4.8 will be met.

4.7 Inspection of preservation, packaging, packing, and marking. The inspection of preservation, packaging, packing, and marking shall be in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Transmitters shall be preserved in accordance with MIL-P-116, method II, and packaged in accordance with MIL-STD-794.

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5.1.2 Level C. Transmitters shall be packaged in accordance with MIL-STD-794.

5.2 Packing

5.2.1 Level A. Transmitters shall be packed in accordance with MIL-STD-794.

5.2.2 Level C. Transmitters shall be packed in accordance with MIL-STD-794.

5.3 Marking for shipment. Marking for shipment shall be in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows:

TRANSMITTER, RATE OF FLOW TRU-6/A, 0-5000 PPH

5.3.1 Reinspection date marking. The reinspection date marking shall be in accordance with the instructions of the preparing activity (see 6.2).

6. NOTES

6.1 Intended use. The TRU-6/A transmitter covered by this specification is intended for use in measuring and transmitting a signal proportional to the total mass rate of fuel flow to the main engine of jet aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. That the items specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor' s expense

c. Levels of packaging and packing required and reinspection date marking (see section 5).

6.2.1 Data and hardware. The contractor will submit the following data and hardware to the qualifying activity:

a. Outline drawing of the transmitter showing dimensions, display, et cetera, and all other configuration details required to demonstrate compliance with this specification

b. Internal circuit schematic and interconnecting diagram

c. Installation instruction sheet

d. Three copies of the test report specified in 4.4.2, signed by an Air Force inspector, demonstrating compliance of the transmitter with all the tests specified herein

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e. All qualification test samples in the AS IS condition (to be retained by the procuring activity)

f. When transmitters that have been subjected to the sampling tests (see 4.5.2.2.1) will be submitted to the qualifying activity for inspection prior to reconditioning

g. Test box or wiring harness for laboratory testing.

6.2.2 Reliability test program. Within 90 days after receipt of contract or purchase order and 30 days prior to the scheduled date of initiation of testing or at such later date as the contracting officer may authorize, the contractor will submit a detailed reliability test program to the procuring activity for approval. Testing will begin only after formal approval of the test program by the procuring activity.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to the requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASNFI-20, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 11

Review activities:
Navy - AS
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6620-0239

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD. FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

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